

Week 2 (Thursday, June 26)

Statistical Learning Theory

CMPE-257: Machine Learning - Summer 2025



SJSU SAN JOSÉ STATE
UNIVERSITY

Instructor: Rafael E Banchs
Email: rafael.banchs@sjsu.edu

Today's plan...

Main Topic: Statistical Learning Theory (an Introduction)

5:30-6:45 Session 1

6:45-7:00 Break

7:00-8:45 Session 2

8:45-9:00 Break

9:00-9:30 Term Projects and HW2

Session 1:

- The Learning Problem

Session 2:

- Generalization Theory
- Hands-on coding session 2

First Homework is Due Today!

Week 2 - Session 1

The Learning Problem

- Feasibility of Learning
- Training vs Testing
- Model Complexity and Learning
- Data Complexity and Learning

Recap from last week...



Train Data

Learning

Model



Test Data



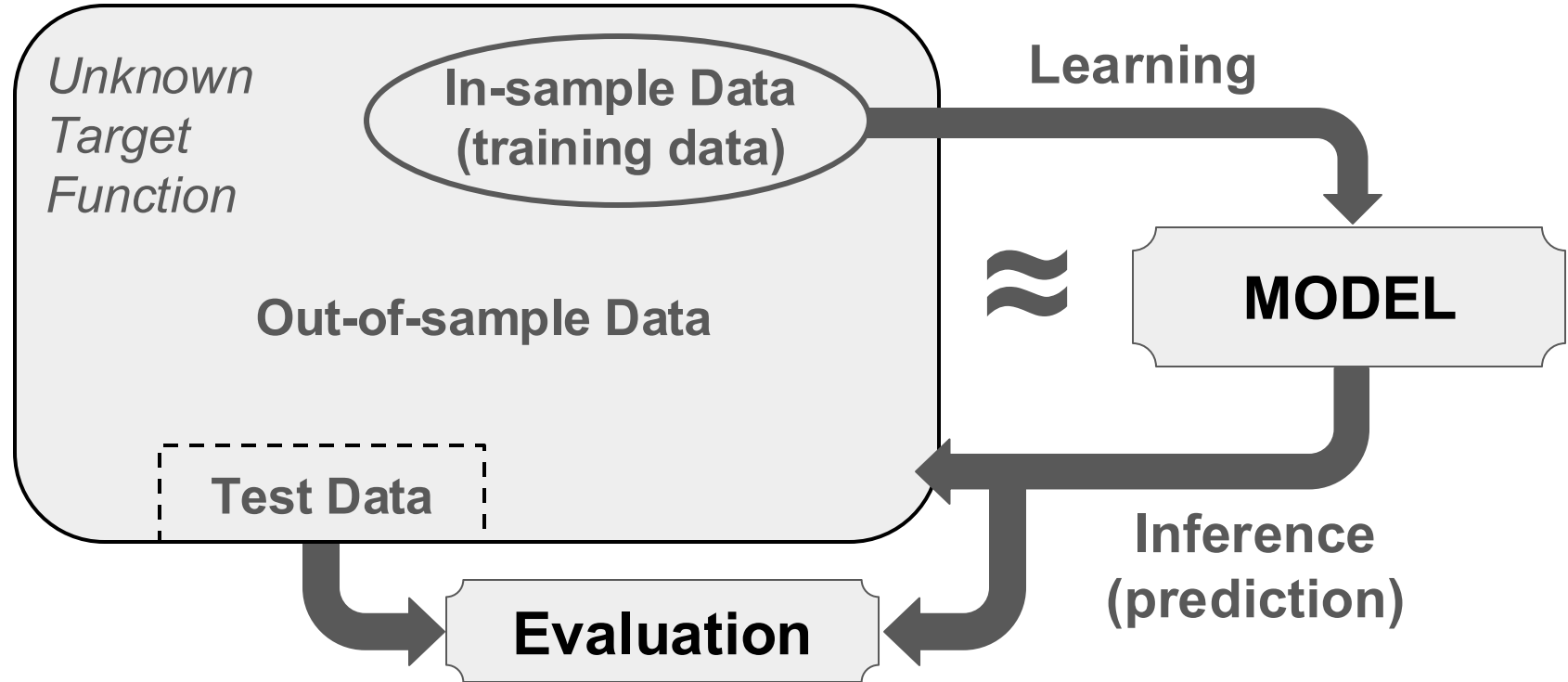
Inference

Definition of the learning problem

- For a given set of observations (x_n, y_n) for $n = 1, 2, \dots, N$, which are produced by an unknown function $y = f(x)$
- Can we find a function $g(x)$ such that $g(x) \approx f(x)$ for all x , even for those (x_k, y_k) outside the set of observations?
- In general, the answer to this question is “yes”, **but** it depends on whether certain conditions hold or not



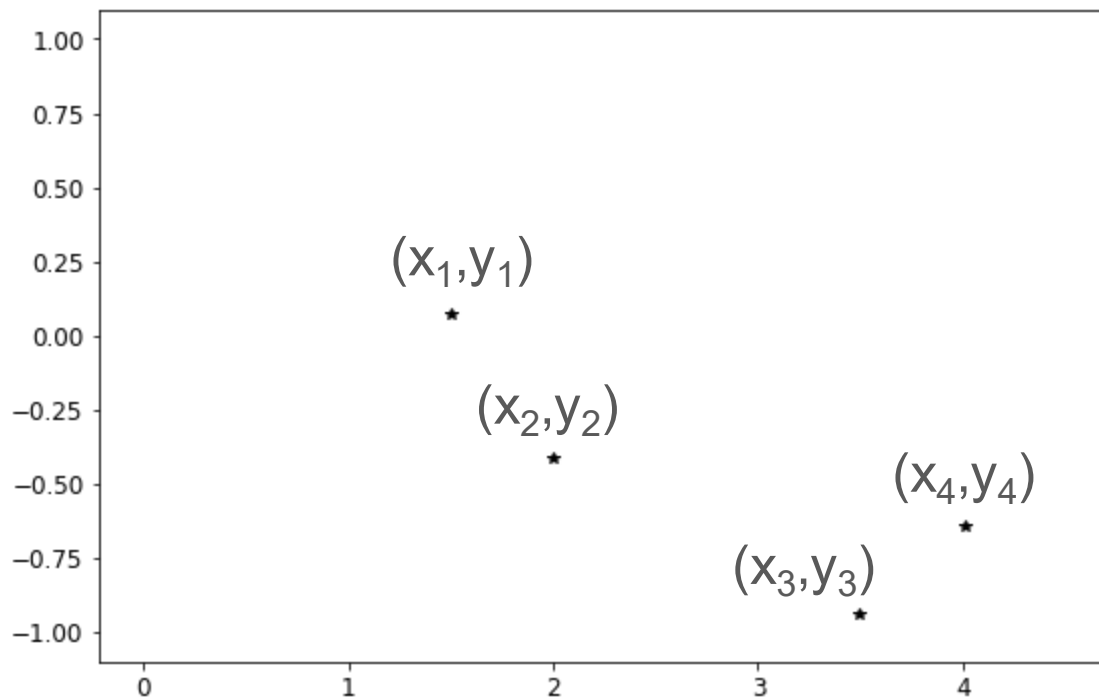
Main components of the learning problem



Important concepts...

- In-sample data: a set of data samples used for learning
- Learning: using an algorithm to learn a model
- Model: a usable representation of the data space
- Inference: using a trained model to analyze new data
- Out-of-sample data: any new data not used for learning
- Evaluation: using new data to assess model performance

An illustrative example...



A **set of points** are given

$$(x_1=1.5012, y_1=0.0695)$$

$$(x_2=1.9981, y_2=-0.4144)$$

$$(x_3=3.4979, y_3=-0.9372)$$

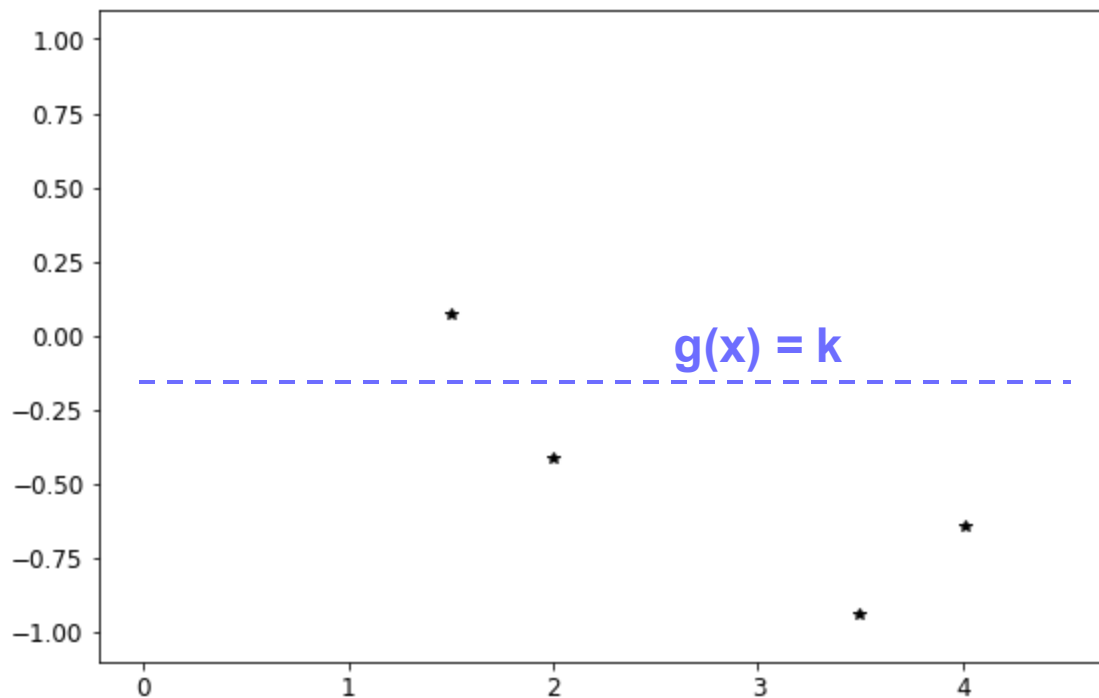
$$(x_4=4.0145, y_4=-0.6426)$$

They come from an

unknown function $f(x)$

Can you find **$g(x)$** such
that $g(x) \approx f(x)$ for all x ?

Case 1: a constant model



Let us start with a very simple model...

$$g(x) = k$$

where k is the only model parameter

How we find the best possible $g(x)$?



Mean Squared Error (MSE)

- We need to measure how close $g(x)$ gets to the actual $f(x)$
- By minimizing such measure we will find the best $g(x)$
- We will see more metrics later, but let us start with MSE

$$\text{MSE} = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2$$

the actual values $y_n = f(x_n)$

the estimates $\hat{y}_n = g(x_n)$



Case 1: optimizing $g(x) = k$

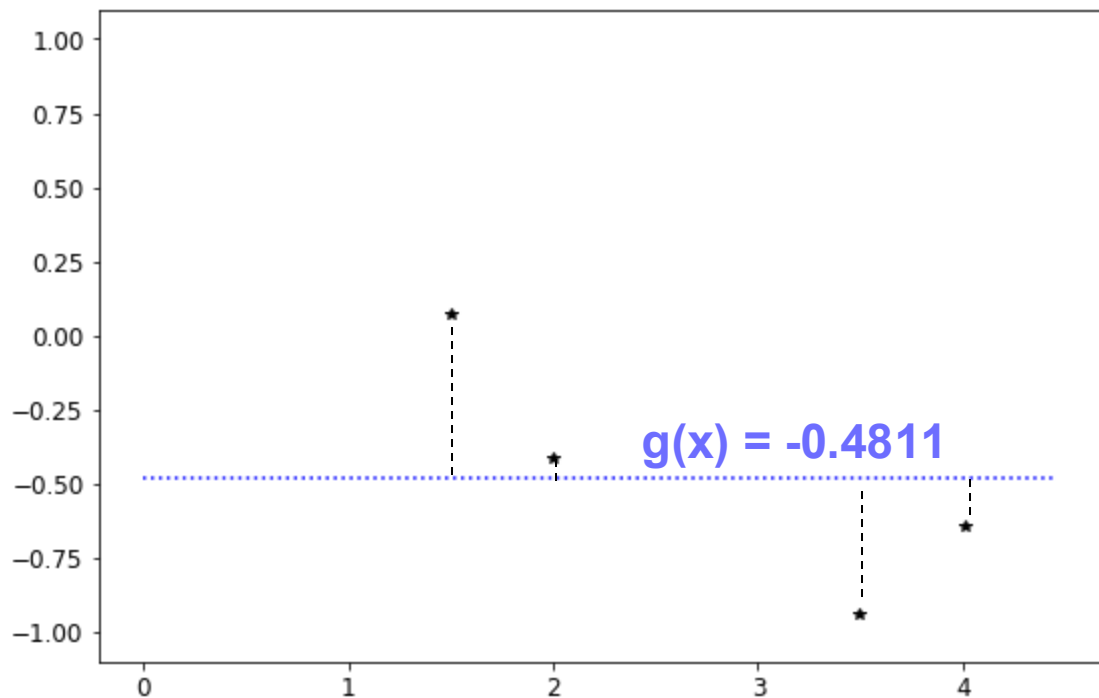
$$\text{MSE} = \frac{1}{4} ((y_1 - k)^2 + (y_2 - k)^2 + (y_3 - k)^2 + (y_4 - k)^2)$$

$$\frac{\delta \text{MSE}}{\delta k} = \frac{1}{2} ((k - y_1) + (k - y_2) + (k - y_3) + (k - y_4)) = \frac{1}{2} (4k - y_1 - y_2 - y_3 - y_4)$$

$$\frac{\delta \text{MSE}}{\delta k} = \frac{1}{2} (4k - y_1 - y_2 - y_3 - y_4) = 0 \implies k = \frac{1}{4} (y_1 + y_2 + y_3 + y_4)$$

The optimal $g(x)$ we are looking for is $g(x) = \frac{1}{N} \sum_n y_n$, $N=4$

Case 1: in-sample (train) MSE



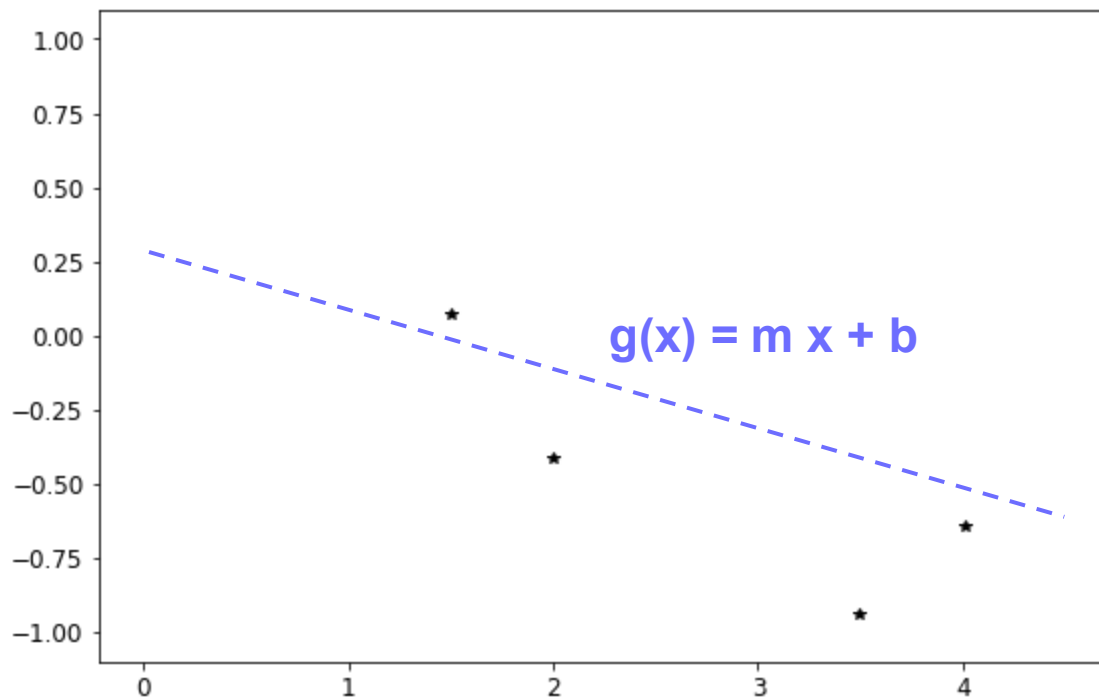
Model:

$$g(x) = k$$

$$k = -0.4811$$

$$\text{MSE} = 0.135438$$

Case 2: a linear model



Let us try now a linear model...

$$g(x) = m x + b$$

where m and b are model parameters

How we find the best possible $g(x)$?



Case 2: optimizing $g(x) = m x + b$

$$\text{MSE} = 1/N \sum_n (y_n - m x_n - b)^2$$

$$\frac{\delta \text{MSE}}{\delta m} = 2/N \sum_n -x_n (y_n - m x_n - b) = 0$$

$$\Rightarrow m \sum_n x_n^2 + b \sum_n x_n = \sum_n x_n y_n \Rightarrow m \sum x^2 + b \sum x = \sum xy$$

$$\frac{\delta \text{MSE}}{\delta b} = 2/N \sum_n -(y_n - m x_n - b) = 0$$

$$\Rightarrow m \sum_n x_n + N b = \sum_n y_n \Rightarrow m \sum x + N b = \sum y$$



Case 2: optimizing $g(x) = m x + b$

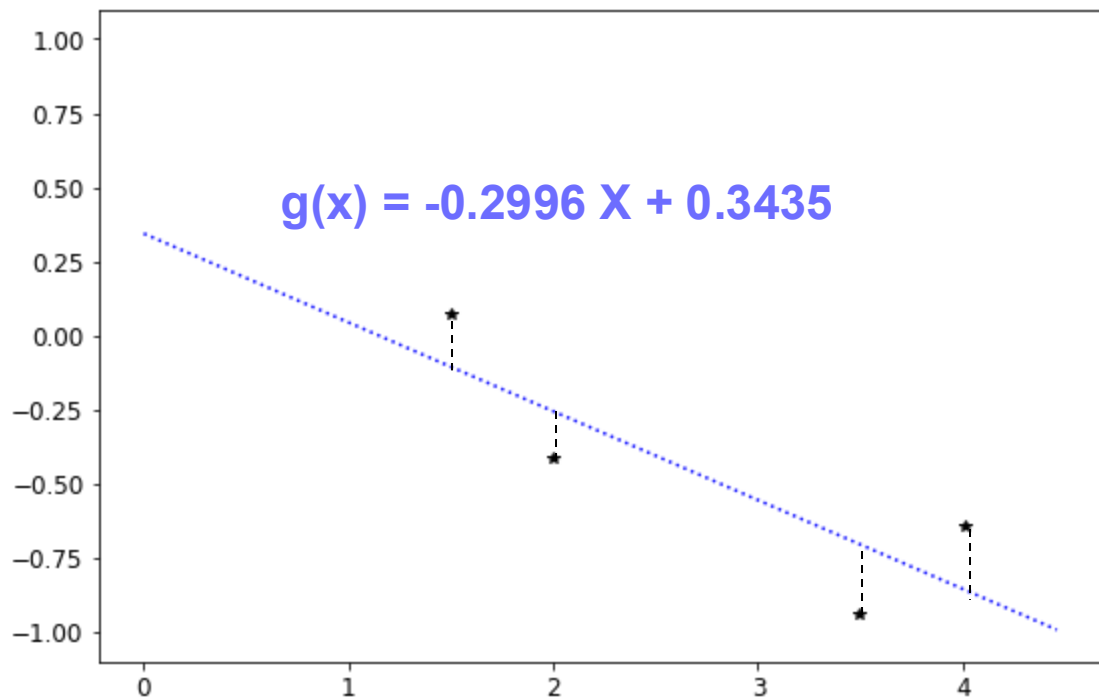
$$m \sum x + Nb = \sum y \Rightarrow b = 1/N (\sum y - m \sum x)$$

$$m \sum x^2 + b \sum x = \sum xy \Rightarrow m \sum x^2 + 1/N (\sum x \sum y - m (\sum x)^2) = \sum xy$$

$$\Rightarrow m N \sum x^2 - m (\sum x)^2 = N \sum xy - \sum x \sum y \Rightarrow m = \frac{N \sum xy - \sum x \sum y}{N \sum x^2 - (\sum x)^2}$$

$$\text{The optimal model is } g(x) = \frac{4 \sum xy - \sum x \sum y}{4 \sum x^2 - (\sum x)^2} (x - \frac{1}{4} \sum x) + \frac{1}{4} \sum y$$

Case 2: in-sample (train) MSE



Model:

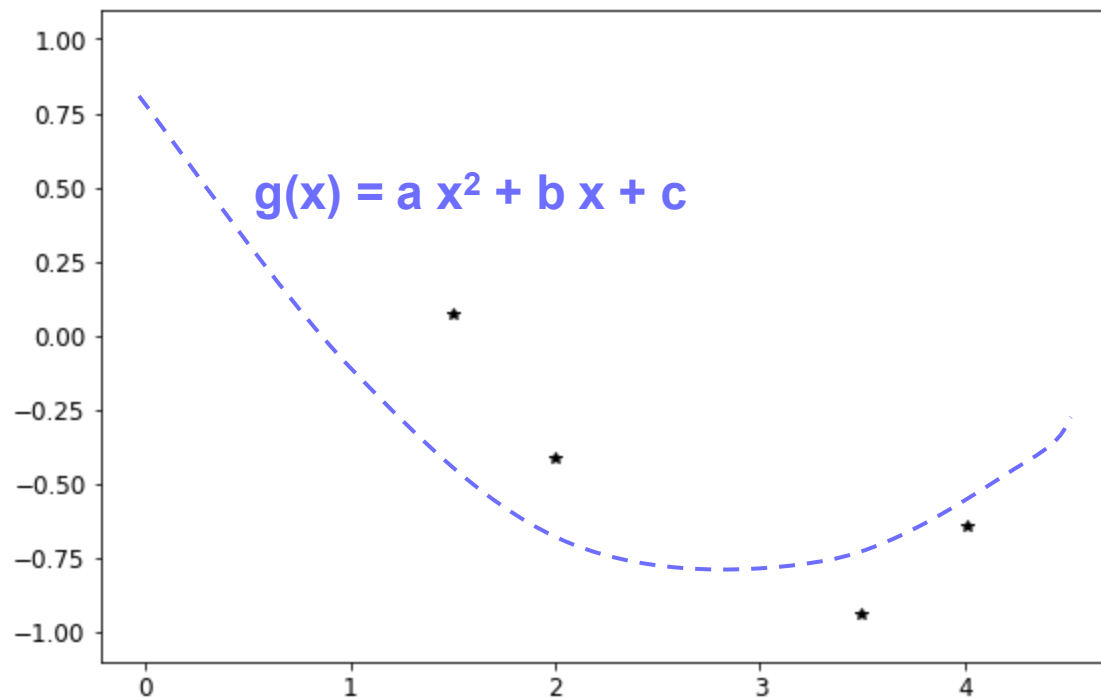
$$g(x) = m x + b$$

$$m = -0.2996$$

$$b = 0.3435$$

$$\text{MSE} = 0.039343$$

Case 3: a quadratic model



Let us try now a quadratic model...

$$g(x) = a x^2 + b x + c$$

where a , b and c are model parameters

How we find the best possible $g(x)$?



Case 3: optimizing $g(x) = a x^2 + b x + c$

$$\text{MSE} = 1/N \sum_n (y_n - a x_n^2 - b x_n - c)^2$$

$$\frac{\delta \text{MSE}}{\delta a, b, c} = (0, 0, 0)$$

$$\Rightarrow c = 1/N (\sum y - a \sum x^2 - b \sum x)$$

$$\sum y x^k = \sum_n y_n x_n^k$$

$$\sum x^k = \sum_n x_n^k$$

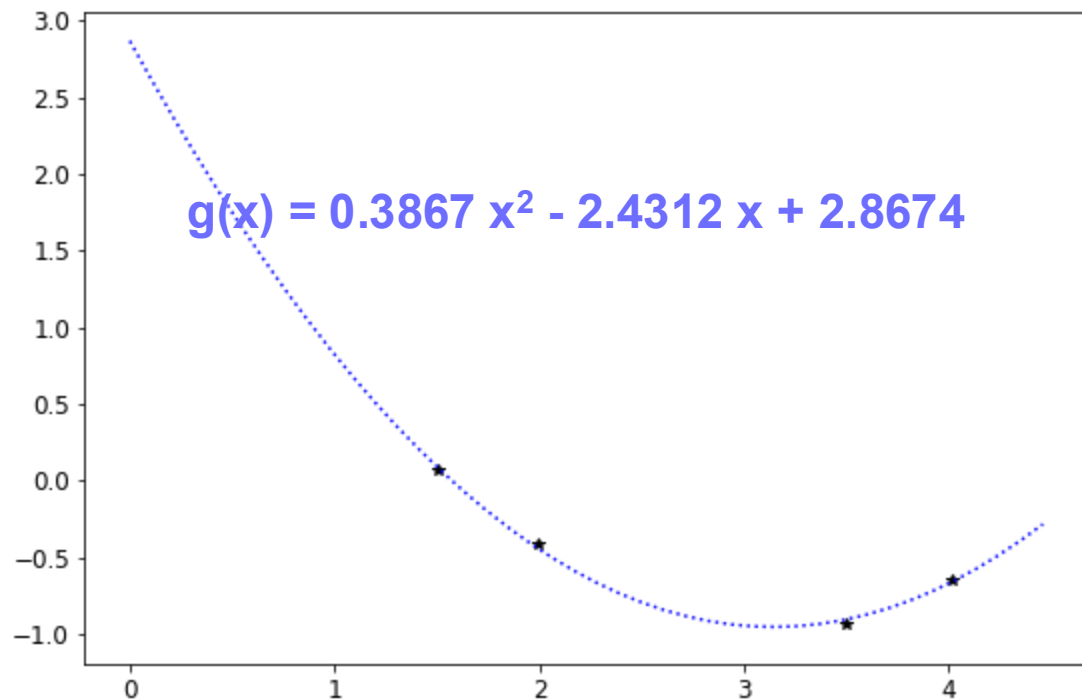
$$\Rightarrow a = \frac{N \sum y x - \sum y \sum x - b (N \sum x^2 - (\sum x)^2)}{N \sum x^3 - \sum x \sum x^2}$$

$$\sum y = \sum_n y_n$$

$$N = 4$$

$$\Rightarrow b = \frac{(N \sum y x^2 - \sum y \sum x^2) (N \sum x^3 - \sum x \sum x^2) - (N \sum y x - \sum y \sum x) (N \sum x^4 - (\sum x^2)^2)}{(N \sum x^3 - \sum x \sum x^2)^2 - (N \sum x^2 - (\sum x)^2) (N \sum x^4 - (\sum x^2)^2)}$$

Case 3: in-sample (train) MSE



Model:

$$g(x) = a x^2 + b x + c$$

$$a = 0.3867$$

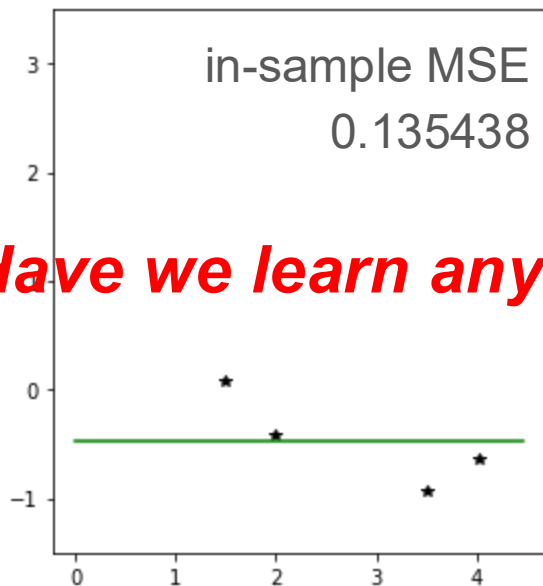
$$b = -2.4312$$

$$c = 2.8674$$

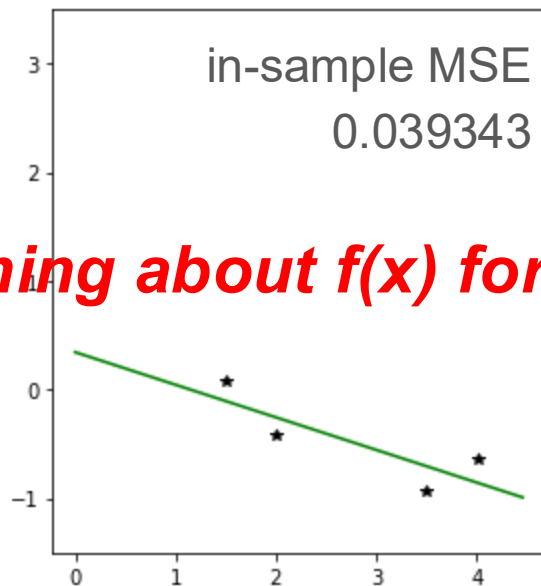
$$\text{MSE} = 0.000688$$

In-sample (train) MSE comparative analysis

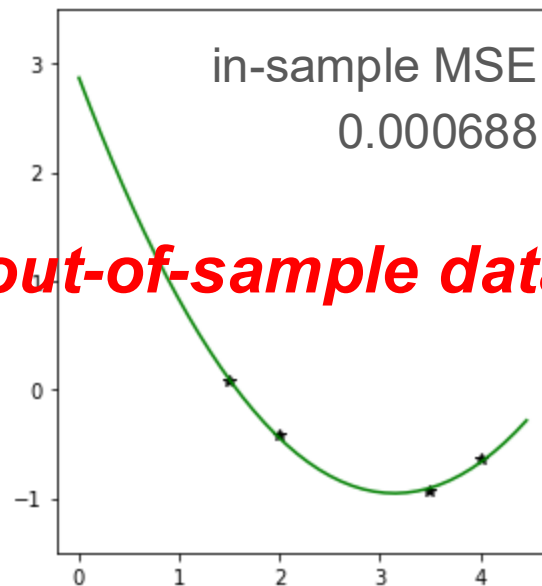
Constant Model



Linear Model

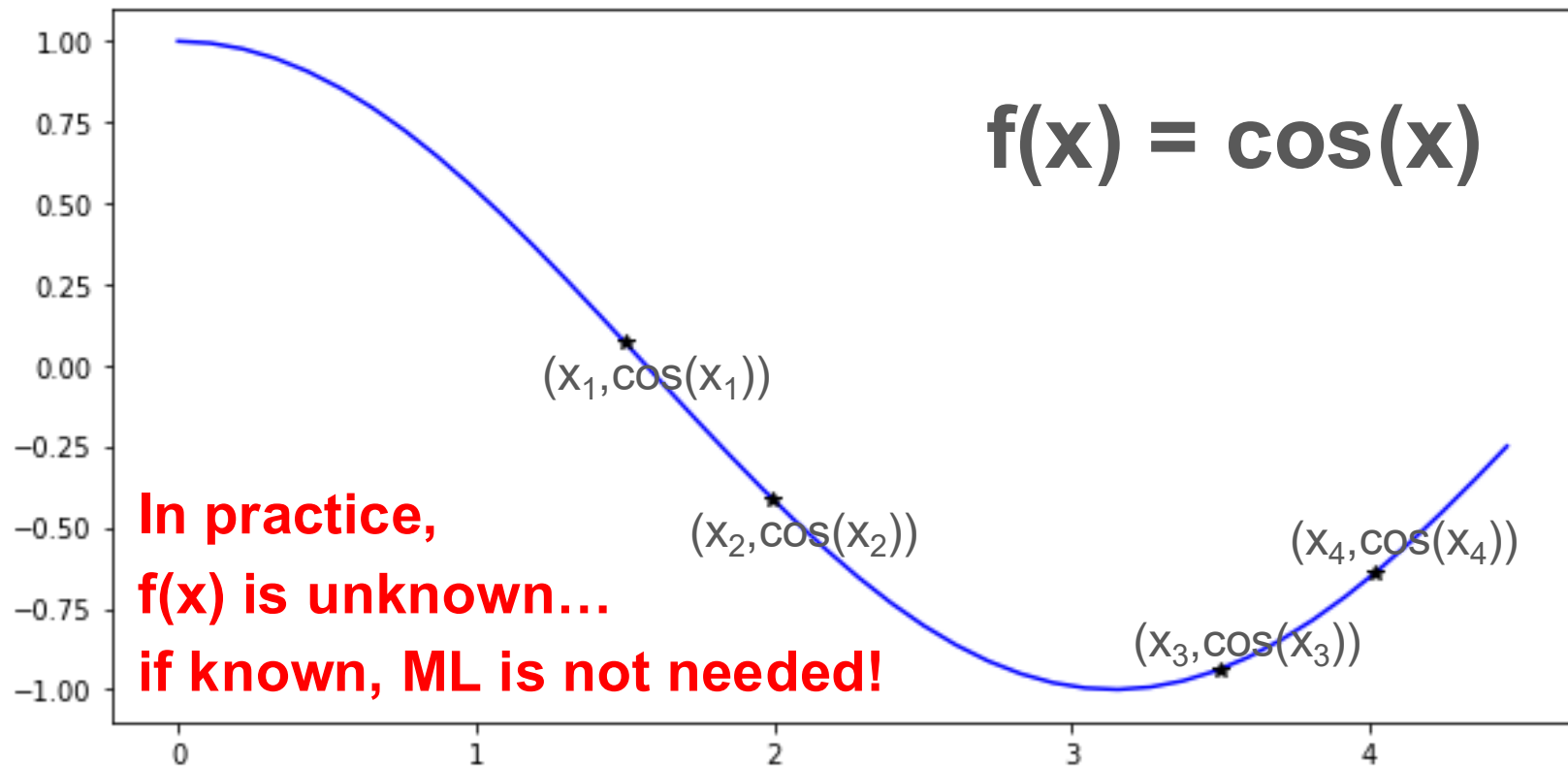


Quadratic Model



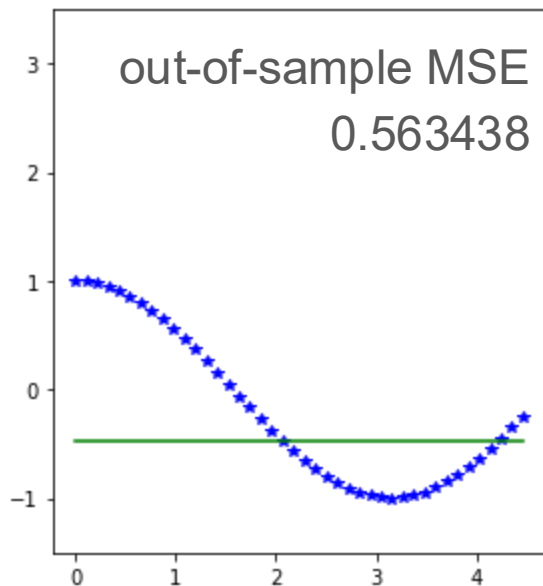
Have we learn anything about $f(x)$ for out-of-sample data?

The actual function behind the given data points...

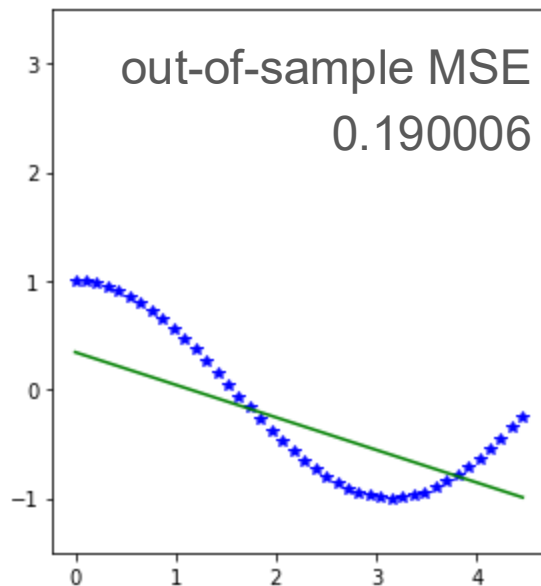


Out-of-sample (test) MSE comparative analysis

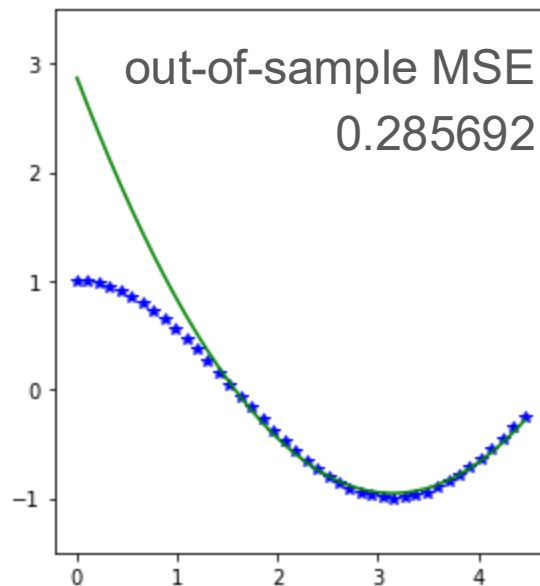
Constant Model



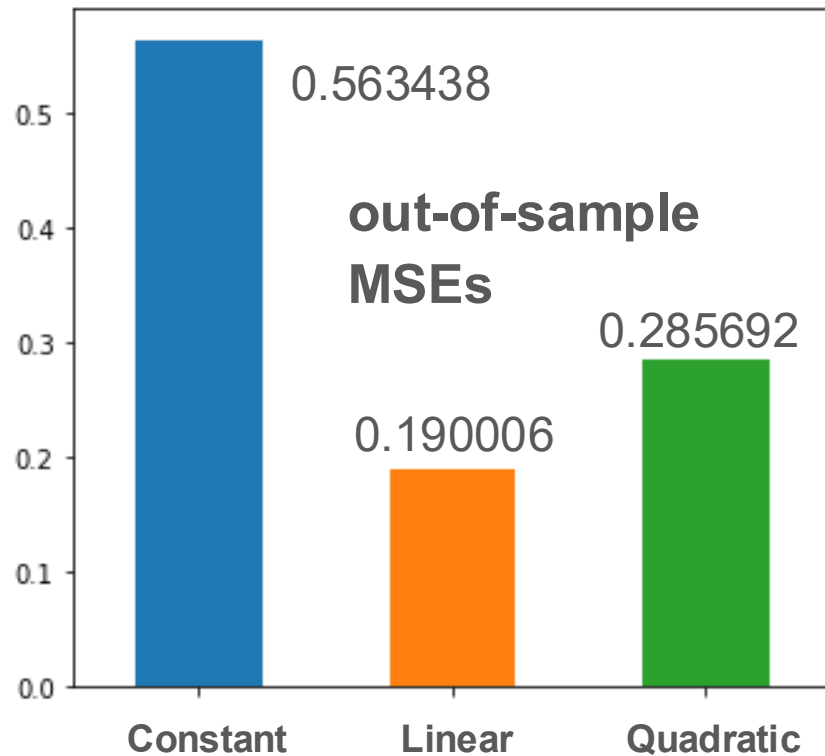
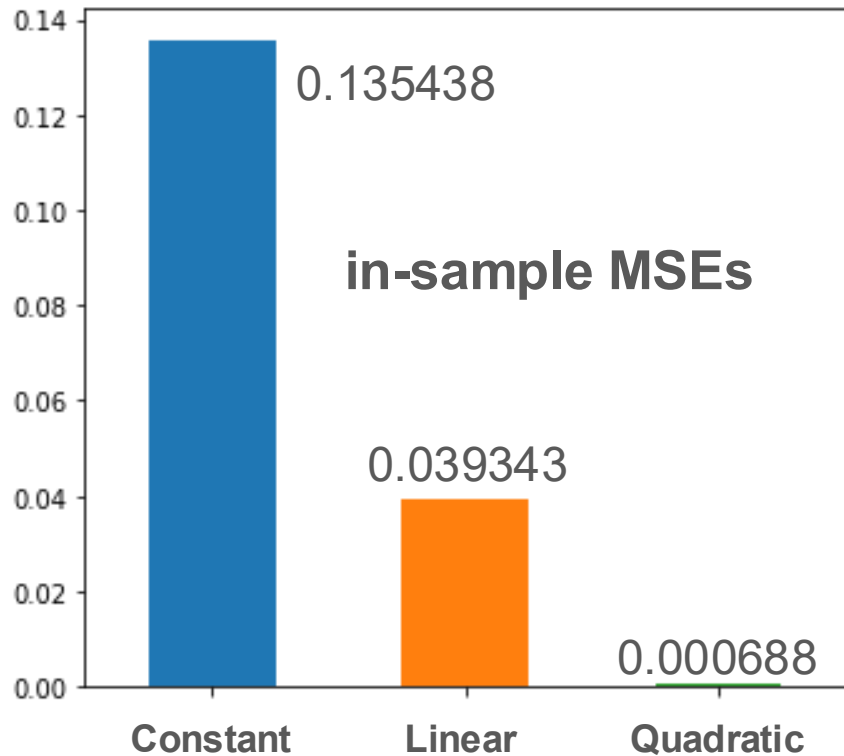
Linear Model



Quadratic Model



In-sample (train) vs out-of-sample (test) MSE



Observations on model performances

- Constant: the worst in both for in-sample and out-of-sample
- Linear: the best for out-of-sample but not for in-sample
- Quadratic: the best for in-sample but not for out-of-sample

Conclusion...

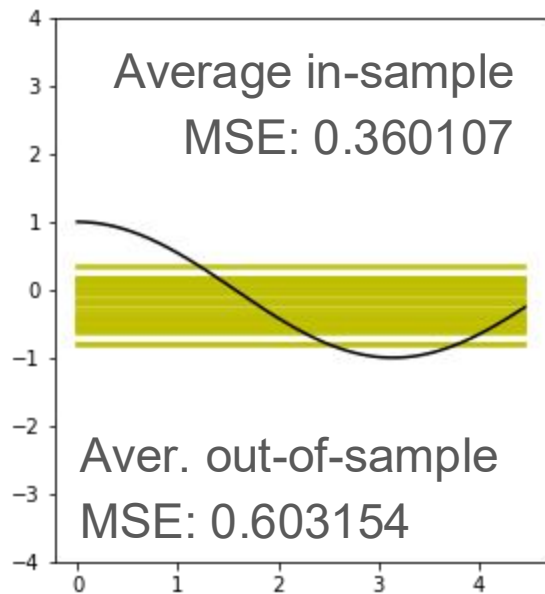
- Best in-sample fit does not guarantees best out-of-sample fit
- Can we conclude anything about what is the best $g(x)$ to approximate $f(x)$ overall? **NO WITH A SINGLE EXPERIMENT**

Average in-sample and out-of-sample MSEs

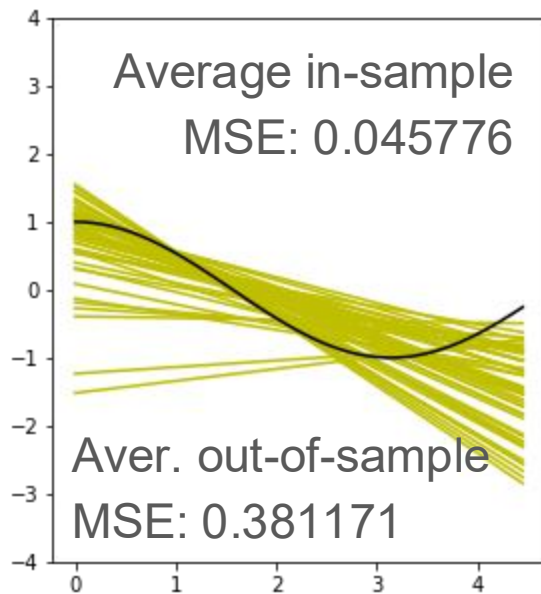
- Consider a large number N of experiments ($N=10K$)
- For each different experiment:
 - Randomly selected an in-sample dataset $\{p_1, p_2, p_3, p_4\}$
 - Get best model fit $g_n(x)$ for the in-sample dataset
 - Compute both in-sample and out-of-sample MSEs
- Compute averages of in-sample and a out-of-sample MSEs

The first fifty models for each model type in the simulation

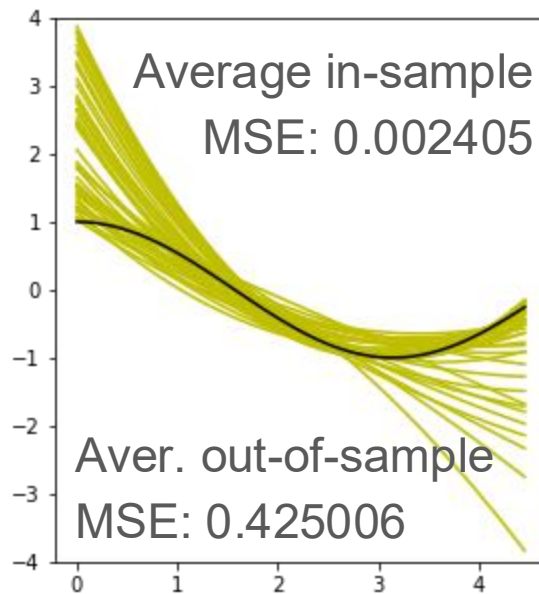
Constant Model



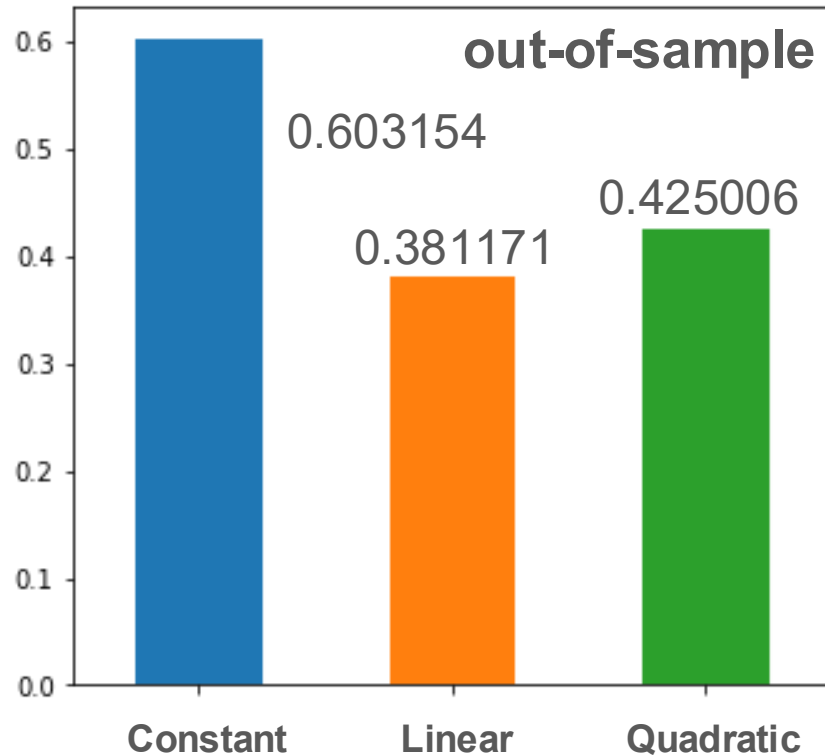
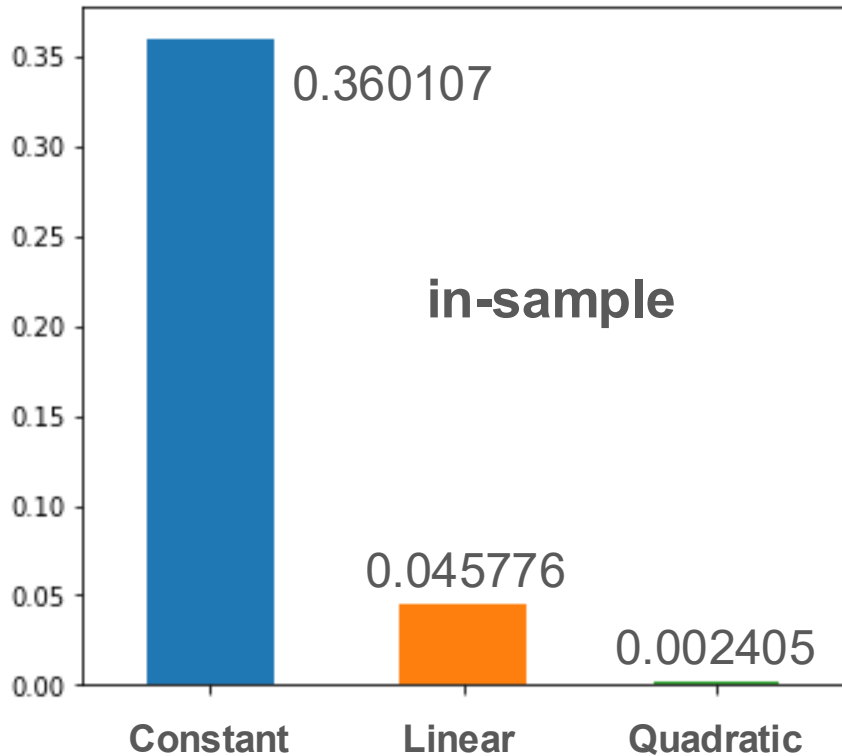
Linear Model



Quadratic Model



Average in-sample (train) and out-of-sample (test) MSEs



Observations on average model performances

- On average, the models perform similarly as before
- Linear exhibits better out-of-sample MSE than quadratic
- However, such difference is not as pronounced as before

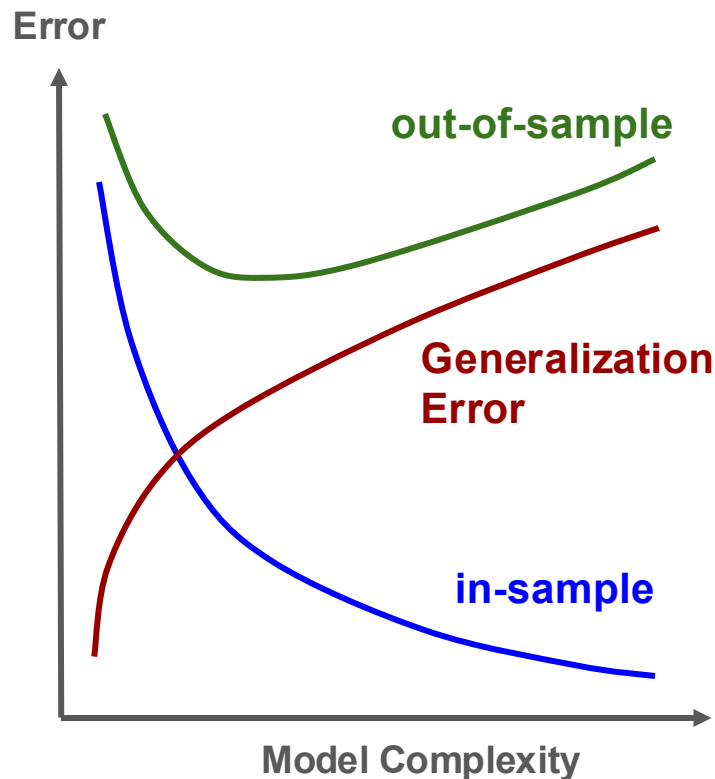
Conclusion...

- Best in-sample fit does not guarantees best out-of-sample fit
- Can we conclude anything about what is the best $g(x)$ to approximate $f(x)$ overall? **YES... ON AVERAGE!**

Model complexity and learning

- Model complexity helps in reducing the in-sample error
- There a price to pay (penalty) in terms of out-of-sample error
- The optimal model is always a compromise between model complexity and in-sample error minimization

$$E_{\text{OUT}} \approx E_{\text{IN}} + \text{GenE}$$



An alternative view... model bias and variance

- Again, for a larger number of experiments...
- A distribution of models is obtained: $g_n(x)$ for $k=1,2,... N$
- Consider the expected value for the best fit model $E[g_n(x)]$
 - BIAS*: MSE of average $g_n(x) \Rightarrow \text{BIAS} = (f(x) - E[g_n(x)])^2$
 - VAR: variance of $g_n(x) \Rightarrow \text{VAR} = E[(g_n(x) - E[g_n(x)])^2]$

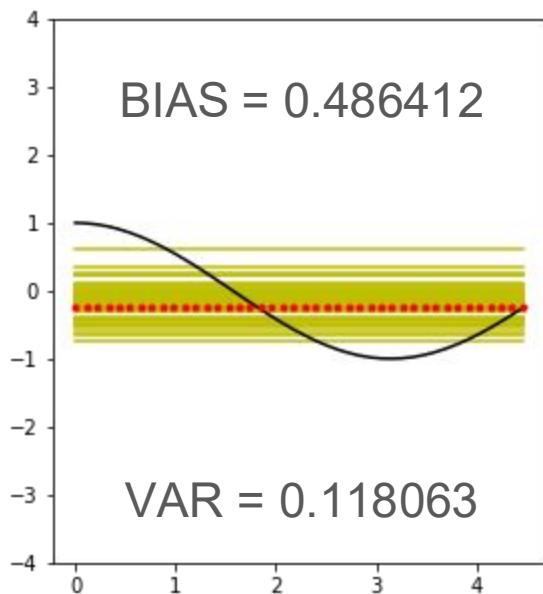
* This bias term is also commonly referred to as the squared-bias

* Notice that these quantities cannot be computed in practice!

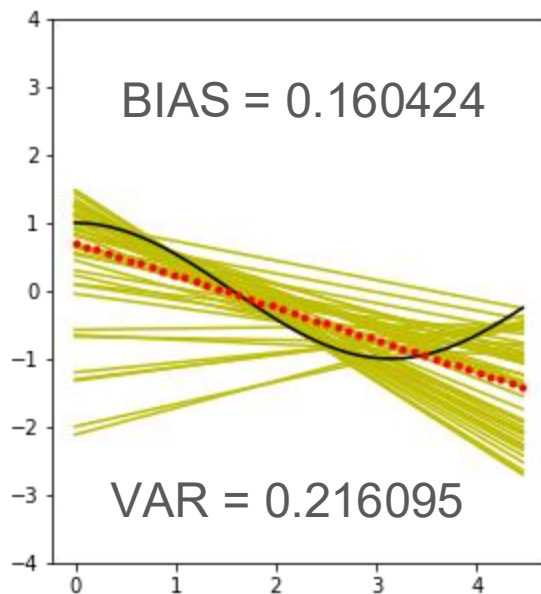
Model bias and variance: visualizations

— $\{g_n(x)\}$ $\cdots E[g_n(x)]$

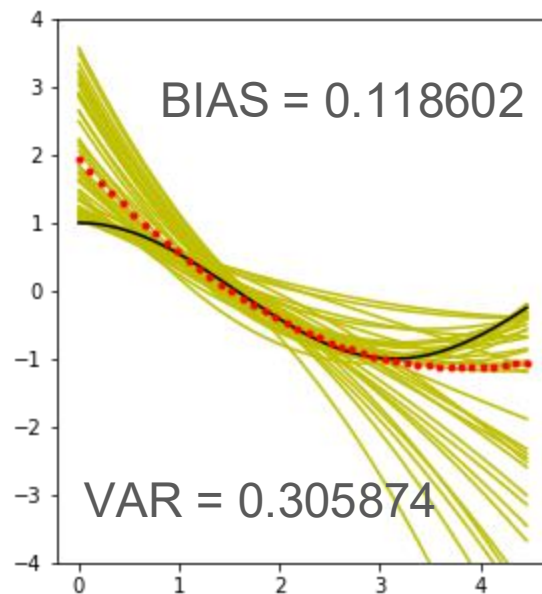
Constant Model



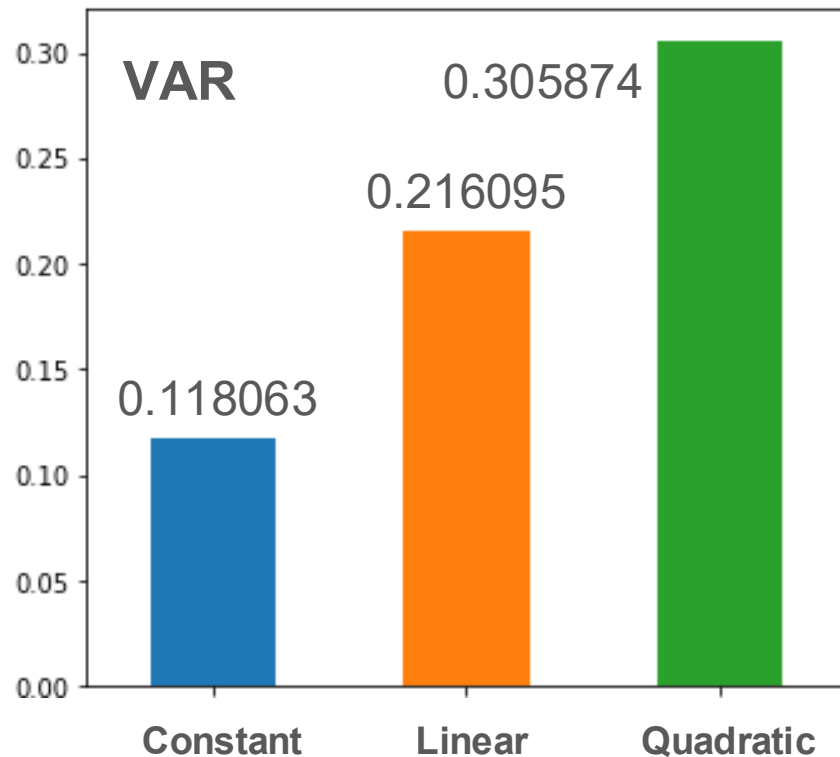
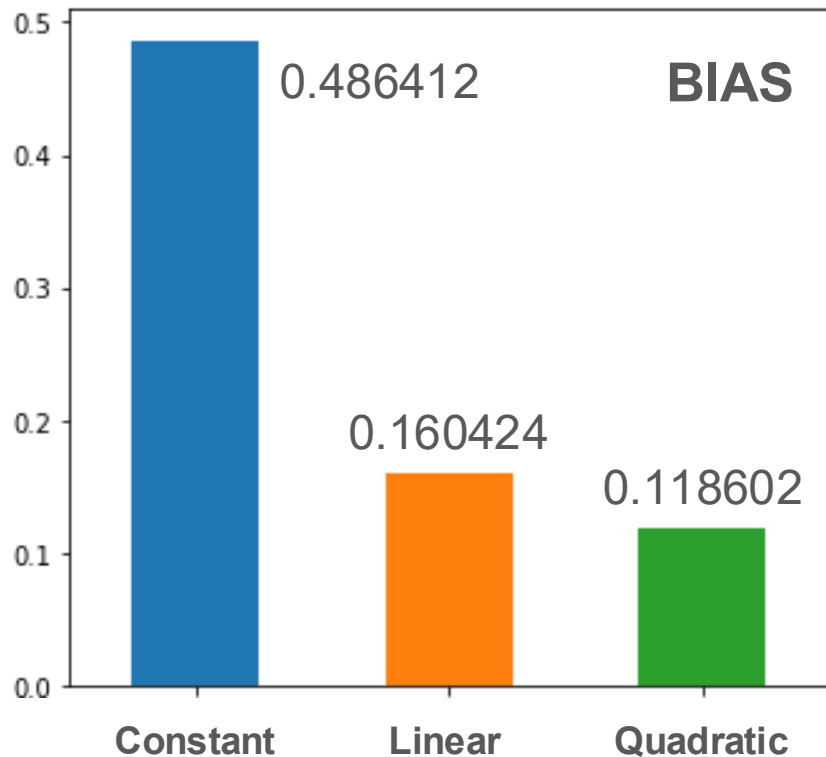
Linear Model



Quadratic Model



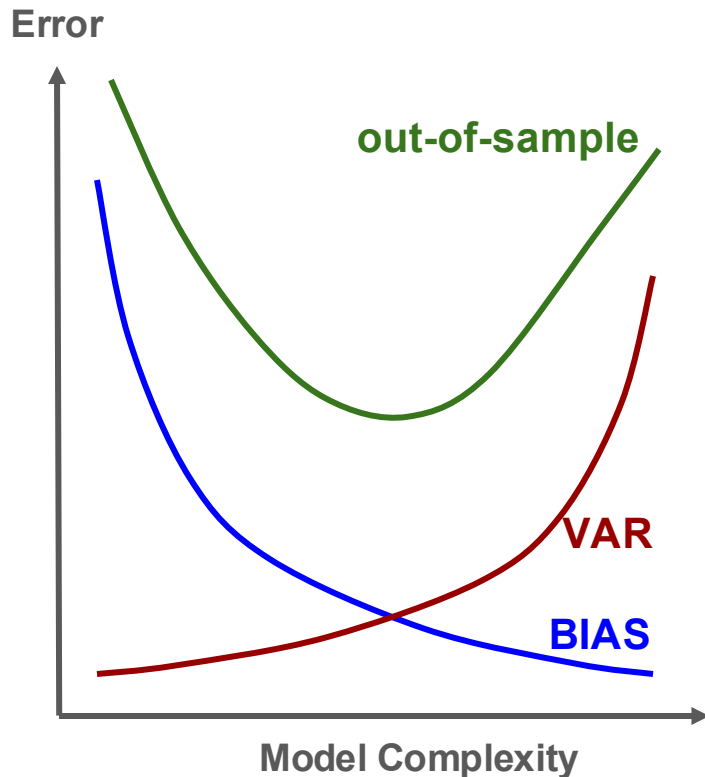
Model bias and variance: comparative analysis



Model BIAS-VAR and learning

- Model complexity helps in reducing the BIAS
- There is a price to pay in terms of VARIANCE
- Ideally, we want to reduce both BIAS and VAR, but optimal models are a compromise between BIAS and VAR

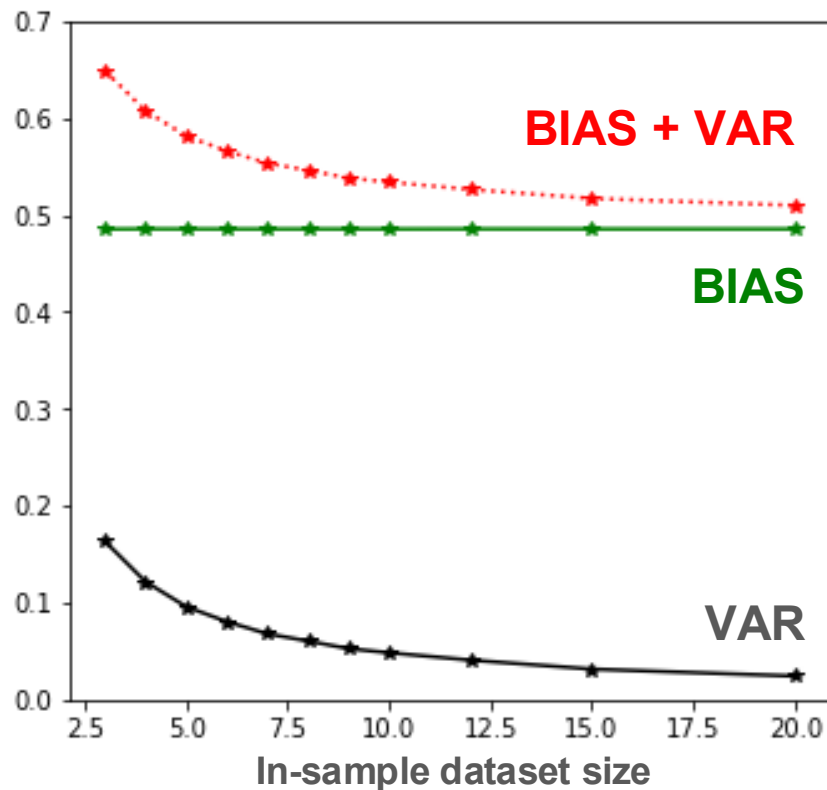
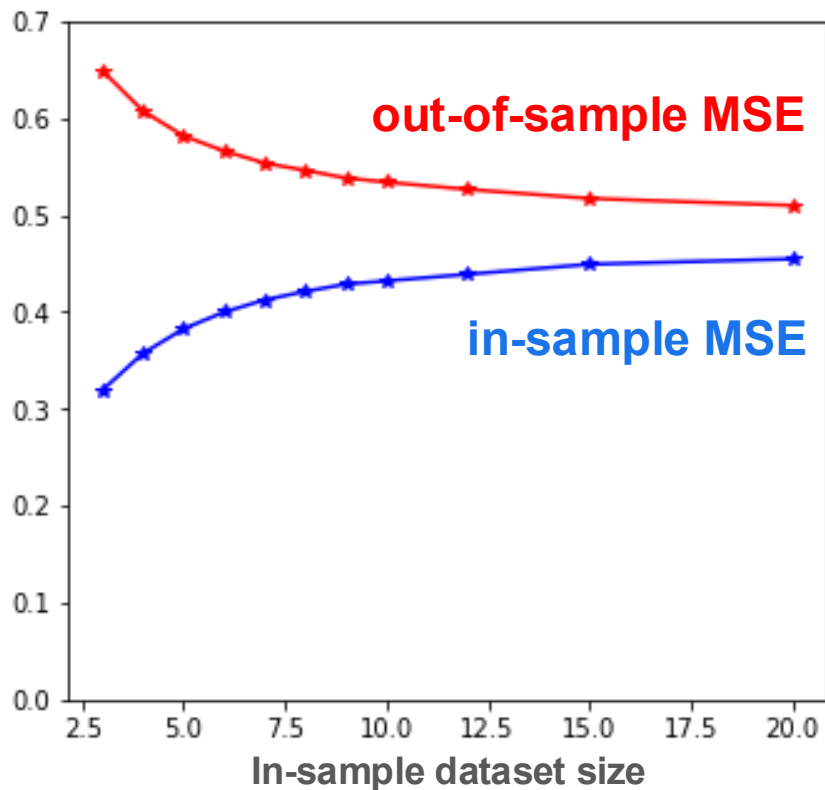
$$E_{\text{OUT}} \approx \text{BIAS} + \text{VAR}$$



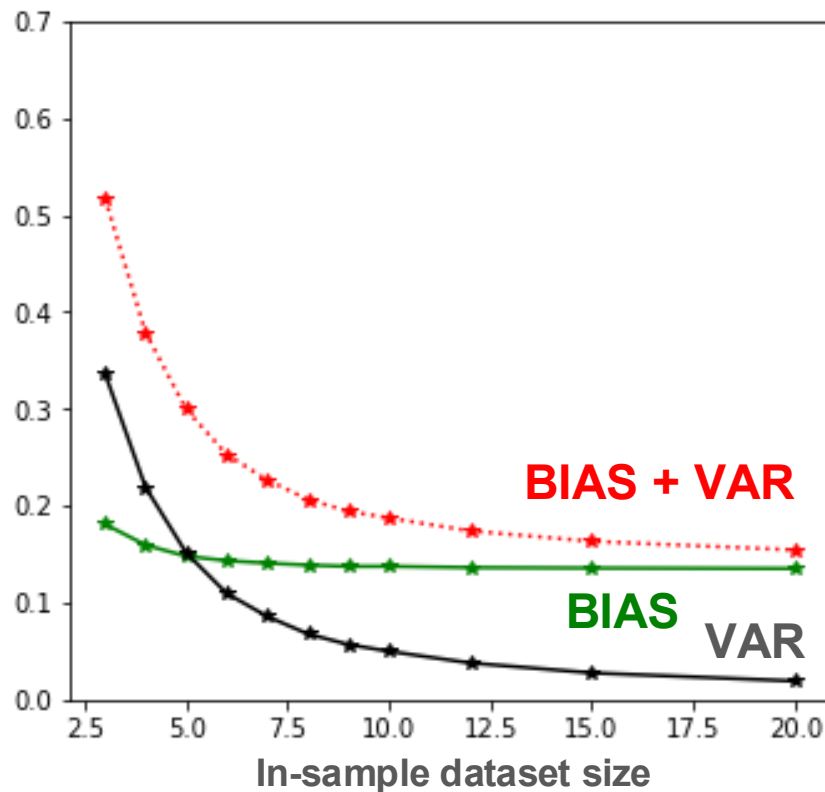
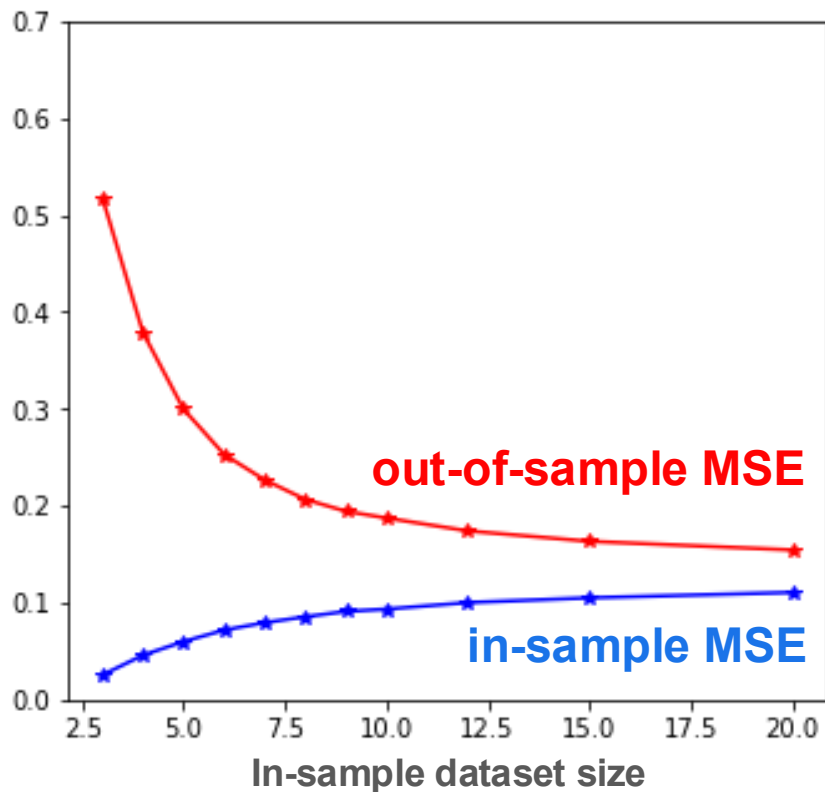
Effects of sample complexity (in-sample dataset size)

- In all previous results, the in-sample dataset size was 4
- In-sample data has strong effects on learning ability:
 - Size: more data samples typically help
 - Quality: in terms of better coverage and representation
- Consider again, larger number N of experiments ($N=10K$)
 - Repeat for different in-sample dataset sizes $\{3,4,5,\dots,20\}$
 - Compute average MSEs, BIAS and VAR for each case

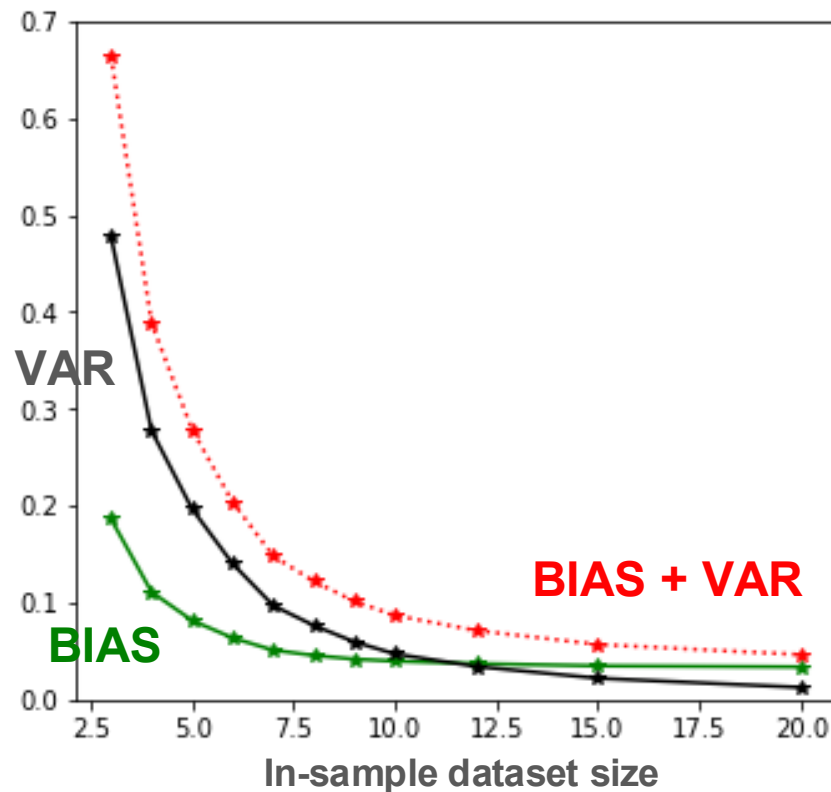
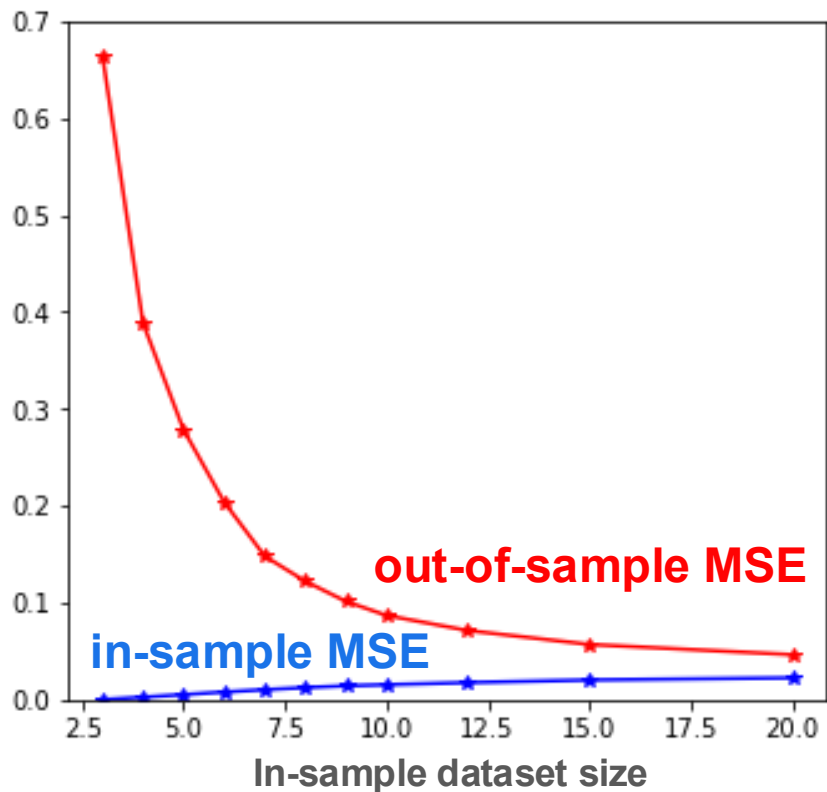
Effects of sample complexity on the constant model...



Effects of sample complexity on the linear model...



Effects of sample complexity on the quadratic model...



Sample complexity and learning

For a fixed model complexity, and increasing sample complexity

Parameter	Behavior	Limit (size $\rightarrow \infty$)
In-sample MSE	increases	BIAS final value
Out-of-sample MSE	decreases	BIAS final value
BIAS	slightly decreases*	BIAS final value
VAR	decreases	Zero

* constant when average $g(x)$ is the best approximation to $f(x)$ for all in-sample sizes

Some important considerations...

- Best in-sample fit does not guarantees best out-of-sample fit
- Higher model complexity is not necessarily better (**we will learn about overfitting and regularization!**)
- Typically, we are not able to compute a closed solution for the best model (**we will learn about learning algorithms!**)
- More in-sample (training) data is typically better
- In practice, the data is typically contaminated with noise!

Noisy data and learning

- In practice, observations are noisy:

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) + \xi \quad \text{where } \xi \text{ is noise!}$$

- Noise can come from different sources:
 - introduced by measuring/observation systems
 - $\mathbf{f}(\mathbf{x})$ is non-deterministic, i.e. probabilistic in nature
 - resulting from additional unknown independent variables



Noisy data and learning

- Assuming a normally distributed noise ξ with zero mean and variance $\text{var}(\xi)$, it can be proven that:

$$\underbrace{E[(f(x) + \xi - g(x))^2]}_{\text{out-of-sample MSE}} = \underbrace{(f(x) - \bar{g}(x))^2}_{\text{BIAS}^*} + \underbrace{E[(g(x) - \bar{g}(x))^2]}_{\text{VAR}} + \underbrace{\text{var}(\xi)}_{\text{Irreducible error}}$$

- The proof is relatively straight forward, but it requires some mathematical juggling. You are encourage to give it a try!

* Reminder: you might see this BIAS term also being referred to as squared-bias



Week 2 - Session 2

Generalization Theory

- Hoeffding Inequality
- VC Dimension
- Generalization Bound

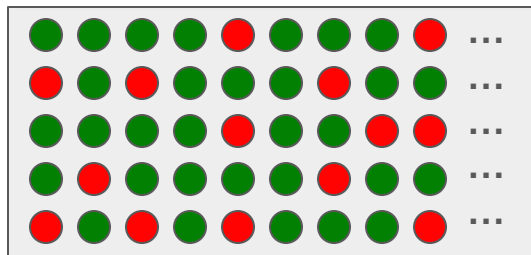
Main learning problem questions

- Q1: Can we make the in-sample error small enough?
 - This can be evaluated after training the model
- Q2: Can we ensure the out-of-sample error is close enough to the in-sample error?
 - This needs to be estimated/validated with test data

In this section, we do a brief incursion into the theory of statistical learning, which deals with the intricacies of Q2



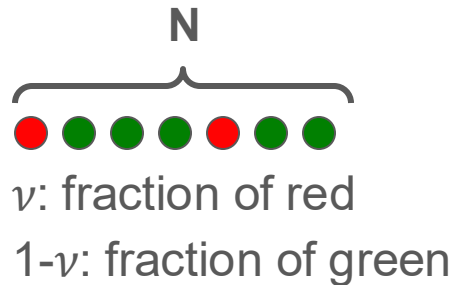
The Hoeffding Inequality



μ : probability of red

$1-\mu$: probability of green

random sampling
independent, with replacement



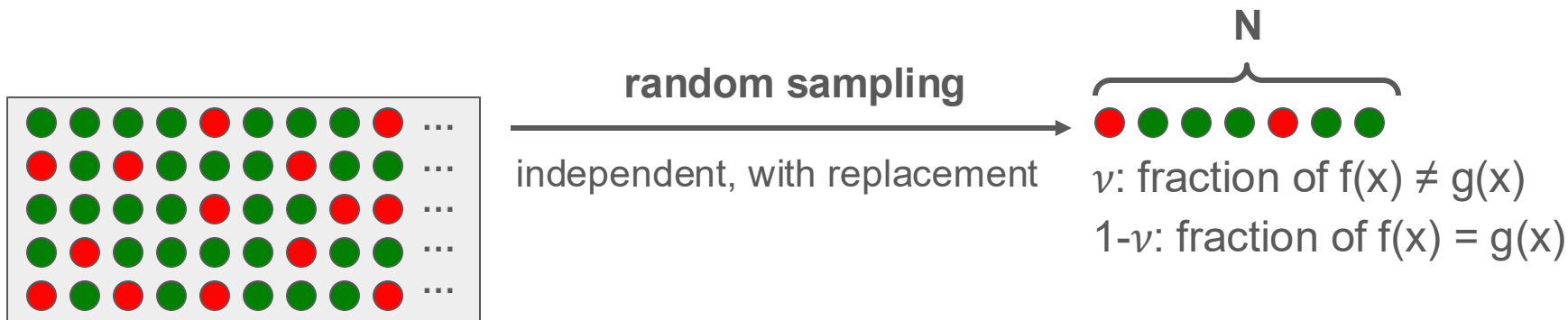
for a given error $\epsilon > 0$

$$P(|\nu - \mu| > \epsilon) \leq 2 \exp(-2\epsilon^2 N)$$

As the sample size N grows the probability of ν deviating from μ gets smaller!

Important: ν is random and depends on N , while μ is not random but it is unknown

The Hoeffding Inequality and the learning problem



μ : probability of $f(x) \neq g(x)$
 $1-\mu$: probability of $f(x) = g(x)$

μ : out-of-sample error $\rightarrow E_{\text{out}}(g)$

ν : in-sample error $\rightarrow E_{\text{in}}(g)$

for a given error $\epsilon > 0$

$$P(| E_{\text{in}}(g) - E_{\text{out}}(g) | > \epsilon) \leq 2 \exp(-2\epsilon^2 N)$$

Limitation of the Hoeffding Inequality

When applying the inequality to the learning problem:

- $E_{\text{in}}(g)$ is a random variable that depends on both the in-sample dataset and the hypothesis g
- $E_{\text{out}}(g)$ is an unknown non-random value that depends on the hypothesis g
- The hypothesis g has to be selected before generating the in-sample dataset for the inequality to hold
- But... learning is about deriving a good g from in-sample data!



Probability rules refresh...

Two important probability rules to be used next:

- If event A implies event B, then the probability of A is less than or equal to the probability of B

$$\text{if } A \Rightarrow B, \text{ then } P(A) \leq P(B)$$

- The probability of the union of events is less than or equal to the sum of the event probabilities (Boole's inequality)

$$P(A \text{ or } B \text{ or } \dots F) \leq P(A) + P(B) + \dots P(F)$$

Hoeffding Inequality with multiple hypothesis g_k

We need to extend the inequality to multiple hypothesis:

- Suppose a finite number of hypothesis $\{g_1, g_2, g_3, \dots, g_M\}$
- From which we want to select a good hypothesis g^*
- Since g^* is one of the hypothesis in the set, we can claim that:

$$P(|E_{\text{in}}(g^*) - E_{\text{out}}(g^*)| > \epsilon) \leq P(|E_{\text{in}}(g_1) - E_{\text{out}}(g_1)| > \epsilon \text{ OR } |E_{\text{in}}(g_2) - E_{\text{out}}(g_2)| > \epsilon \text{ OR } \dots |E_{\text{in}}(g_M) - E_{\text{out}}(g_M)| > \epsilon) \leq \sum_{m=1}^M P(|E_{\text{in}}(g_m) - E_{\text{out}}(g_m)| > \epsilon)$$

Hoeffding Inequality with multiple hypothesis g_k

The final form of the inequality for multiple hypothesis is:

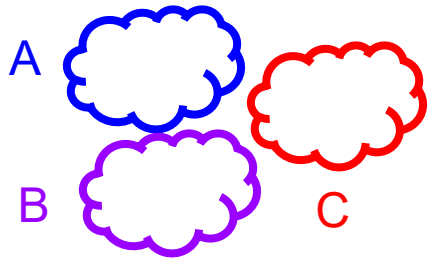
$$\mathbf{P}(| E_{\text{in}}(g^*) - E_{\text{out}}(g^*) | > \epsilon) \leq \mathbf{2M \exp(-2\epsilon^2 N)}$$

- This provides a uniform estimate as it simultaneously approximates all $E_{\text{out}}(g_k)$ by their corresponding $E_{\text{in}}(g_k)$
- The resulting probability bound is too loose to be useful in practice (i.e. consider $M \rightarrow \infty$, an infinite hypothesis set)
- A tighter probability bound is needed!



Probabilities of the union of events revised

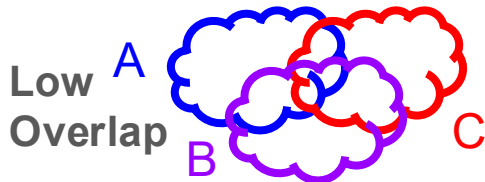
$$P(A \cup B \cup C) \leq P(A) + P(B) + P(C)$$



$$A \cap B \cap C = \emptyset$$

Disjoint Events!

$$\Rightarrow P(A \cup B \cup C) = P(A) + P(B) + P(C)$$



$$A \cap B \cap C \neq \emptyset$$

Overlapping Events!

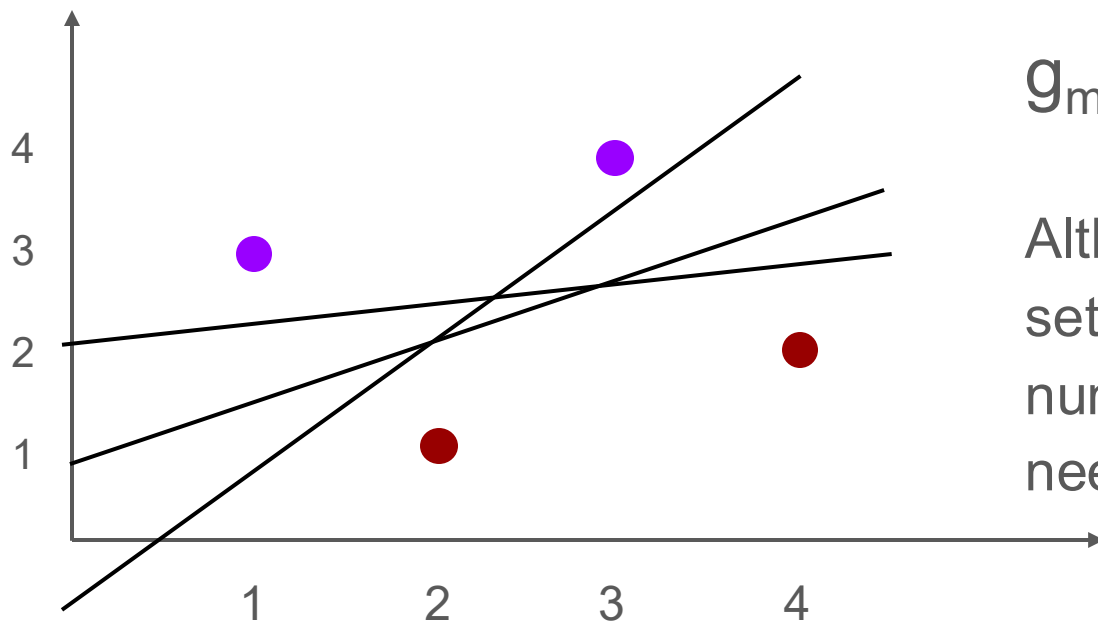
$$\Rightarrow P(A \cup B \cup C) < P(A) + P(B) + P(C)$$



$$\Rightarrow P(A \cup B \cup C) \ll P(A) + P(B) + P(C)$$

Effective number of hypothesis

Consider the classification problem from last week

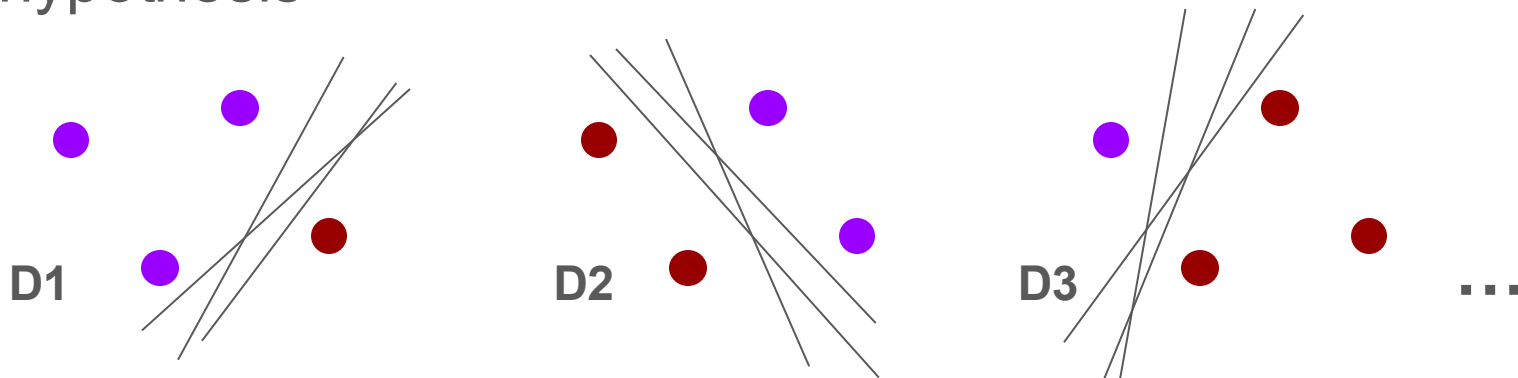


$$g_{m,b}(x) = m x + b$$

Although the hypothesis set is infinite, the effective number of hypothesis we need to select from is not!

The growth function

- $m_G(N)$: the maximum number of dichotomies that can be generated by a hypothesis set G on any N points
- Dichotomy: a common result produced by a group of hypothesis





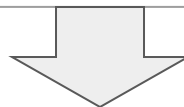
The Vapnik-Chervonenkis (VC) dimension

- dvc: the largest value of N for which $m_G(N) = 2^N$
 - a property of the hypothesis set G
 - a measure of model complexity (degrees of freedom)
 - $m_G(N) \leq N^{\text{dvc}} + 1$  **GOOD MODEL. Learning is possible**
(given a sufficiently large value of N)
 - If $m_G(N) = 2^N$ for all $N \Rightarrow \text{dvc}(G) = \infty$  **BAD MODEL**
No conclusion
can be made!



VC dimension of the linear classifier in $\mathbb{R}^2$

N	1	2	3	4	5
2^N	2	4	8	16	32
$m_G(N)$	2	4	8	14	-



dvc = 3



VC dimension of the linear classifier in $\mathbb{R}^2$

N=1 +///- $m_G(1) = 2$

N=2 ++// +/- -/+ /-- $m_G(2) = 4$

N=3

+ + / + \ - - / + - -
+ / + \ +

+ + + / - - \ + - -
- - - -

$m_G(3) = 8$

N=4

+ + | + \ - - / + - -
+ + + + + + + +

+ + + | - - + - -
+ - + - + - + -

+ + + - - + - -
- + - + - + - +

$m_G(4) = 14$

+ + + \ - - / + - -
- - - - - - - -



The VC generalization bound

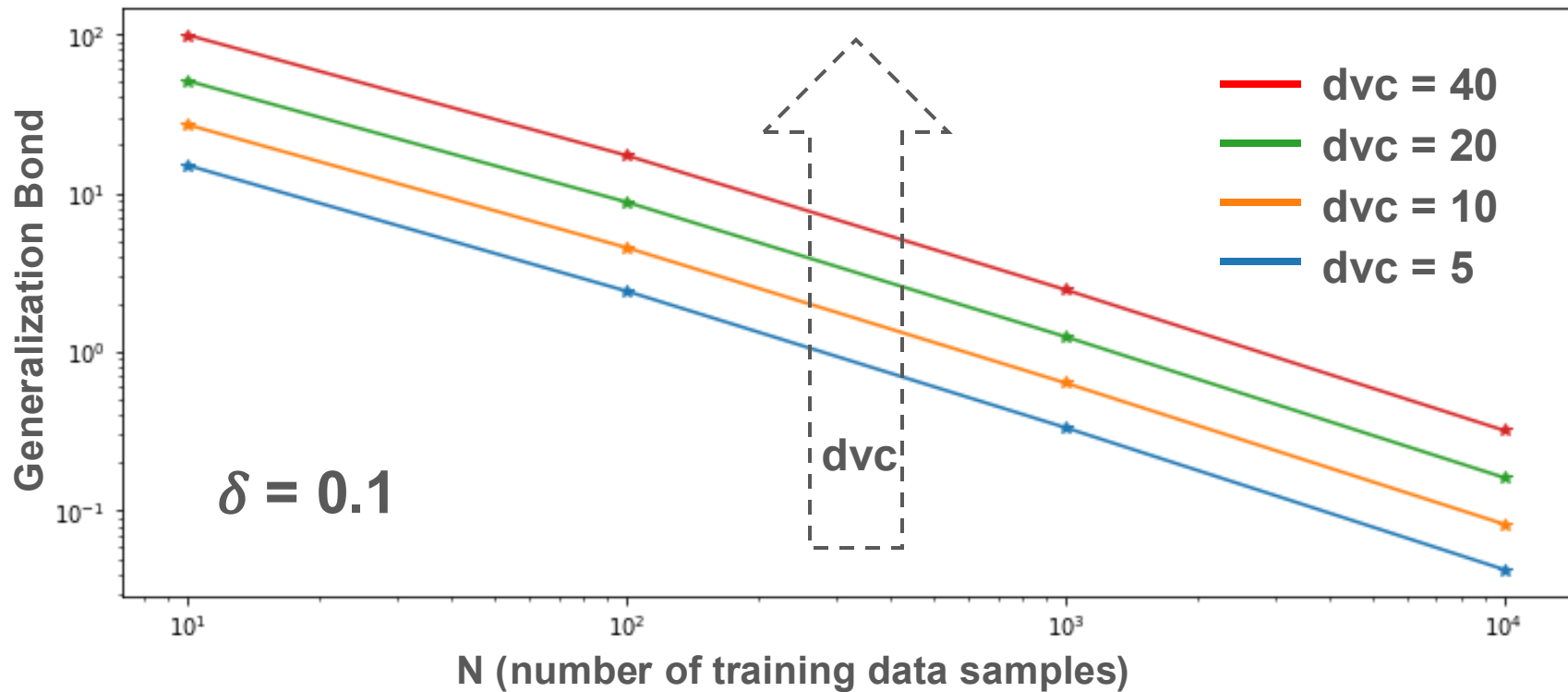
For a given tolerance $\delta > 0$

$$\mathbf{P(E_{out}(g^*) \leq E_{in}(g^*) + GB) \geq 1-\delta}$$

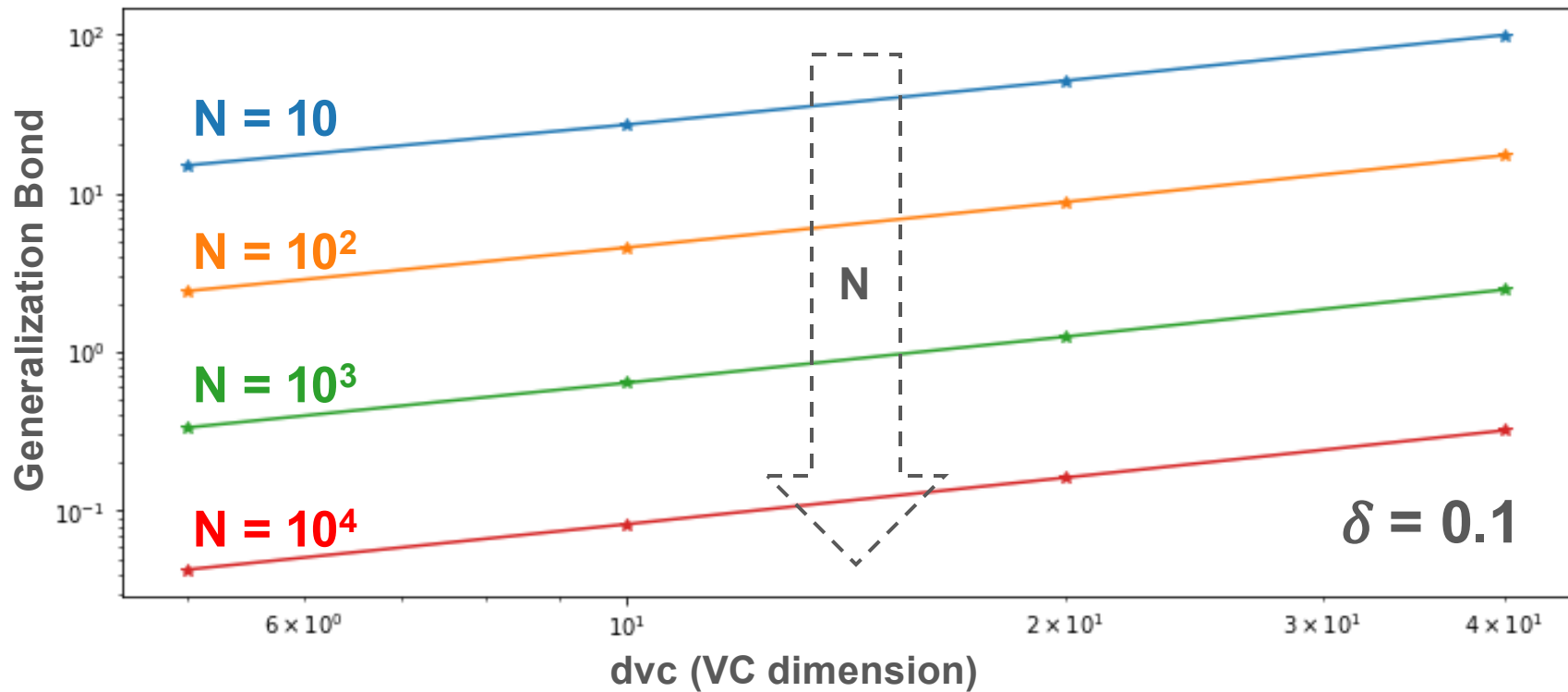
With GB (generalization bound) given by:

$$\mathbf{GB = \sqrt{\frac{8}{N} \ln \left[4 \frac{(2N)^{d_{vc}} + 1}{\delta} \right]}}$$

Generalization bond vs number of training data samples



Generalization bond vs VC dimension





Model complexity exercise 1

- Suppose we are given $N=1000$ data samples and a confidence requirement of 95% (i.e. $\delta = 0.05$)
- How much generalization error is introduced by increasing dvc from 2 to 3?

$$E_{\text{out}}(g^*) \leq E_{\text{in}}(g^*) + \underbrace{\sqrt{\frac{8}{N} \ln \left[4 \frac{(2N)^{\text{dvc}} + 1}{\delta} \right]}}_{\text{Generalization Bound (error)}}$$



Model complexity exercise 1

- $N=1000$, $\delta = 0.05$, $dvc = 2$

$$E_{\text{out}}(g^*) \leq E_{\text{in}}(g^*) + \sqrt{\frac{8}{1000} \ln \left[4 \frac{(2 \times 1000)^2 + 1}{0.05} \right]} \approx E_{\text{in}}(g^*) + 0.3958$$

- $N=1000$, $\delta = 0.05$, $dvc = 3$

$$E_{\text{out}}(g^*) \leq E_{\text{in}}(g^*) + \sqrt{\frac{8}{1000} \ln \left[4 \frac{(2 \times 1000)^3 + 1}{0.05} \right]} \approx E_{\text{in}}(g^*) + 0.4663$$

- $\Delta\epsilon = 0.4663 - 0.3958 = 0.0705$



Sample complexity exercise 2

- Suppose we have a model with $d_{vc} = 2$ and want to achieve a generalization error of 0.1 with confidence of 90%
- How much data do we need?

$$\epsilon \approx \sqrt{\frac{8}{N} \ln \left[4 \frac{(2N)^{d_{vc}} + 1}{\delta} \right]} \Rightarrow N \approx \frac{8}{\epsilon^2} \ln \left[4 \frac{(2N)^{d_{vc}} + 1}{\delta} \right]$$

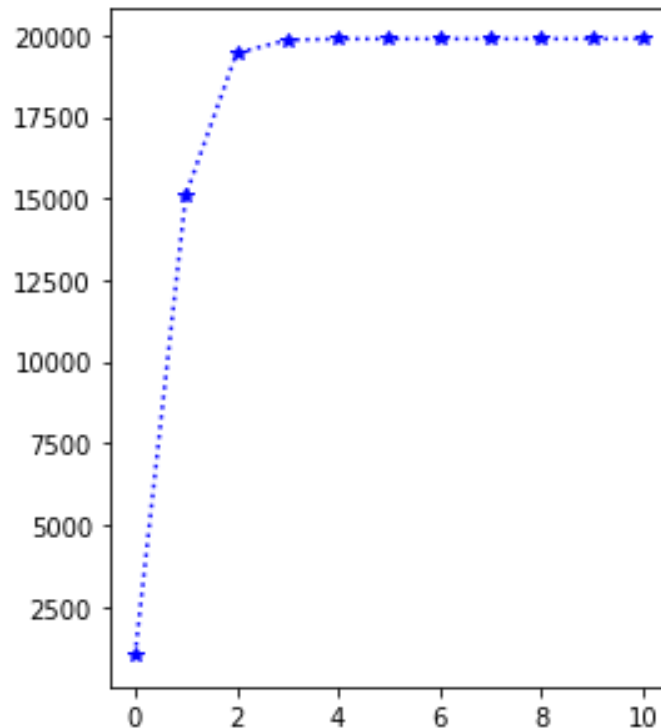


Sample complexity exercise 2

- $dvc = 2, \epsilon = 0.1, \delta = 0.1$
- Start with $N = 1000$ and iterate a few times

$$N \approx \frac{8}{0.01} \ln \left(4 \frac{(2N)^2 + 1}{0.1} \right)$$

- $N \approx 20000$ ($\sim dvc \times 10000$)
- **In practice: $\sim dvc \times 10$**

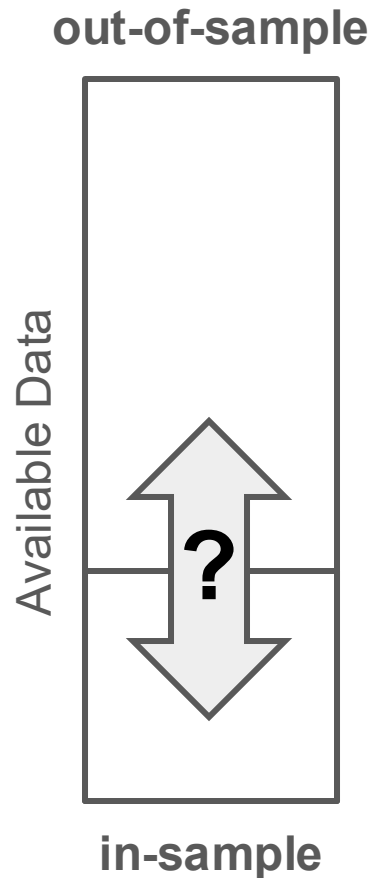


Practical considerations on VC analysis

- With a few exceptions, computing the VC dimension of a model is typically very complex
- The VC generalization bound is still a loose bound
- But it has theoretical value as it shows that learning is possible with infinite hypothesis sets
- Efforts to tighten the VC generalization bound are not of any practical value

Some final considerations...

- In practice:
 - Data is a limited resource
 - BIAS and VAR cannot be computed
 - Actual model complexity is unknown
- We need to be wise on how to use data:
 - In-sample (train) subset
 - Out-of-sample (test) subset





Hands-on Coding Session W2S1

Hands on coding session W2S1



- Look at the notebook and get familiar with its contents
- Reproduce the experimental results presented in class about finding optimal fits (constant, linear, and quadratic) with a single set of four in-sample data (Exercise 1 in PART 1)
- Try to find two other sets of in-sample data, for which the best performing models in terms of out-of-sample MSE are the quadratic and constant models (Exercises 3 and 4 in PART 1)
- Continue working on the rest of the sections in the notebook as permitted by time (HW2 is to complete the entire notebook)

Week 2 - Closing remarks

Wrap-up and Homework 2

- What did we look at today?
- Further reading and support materials
- Second homework assignment
- Term Project Proposal Guidelines

What did we look at today?

The Learning Problem	• Feasibility of Learning • Training vs Testing • Model Complexity and Learning • Data Complexity and Learning
Generalization Theory	• Hoeffding Inequality • VC Dimension • Generalization Bound
Hands-on coding session W2S1	• Approximating an unknown function • Model complexity and learning • Sample complexity and learning • Exploring the effects of noise

Further reading and support materials

- James, Witten, Hastie, Tibshirani, Taylor, “An Introduction to Statistical Learning with Applications in Python” (Chapter 2)
<https://www.statlearning.com/>
- Abu-Mostafa, Magdon-Ismael, Lin, “Learning From Data” (Chapters 1 and 2) <https://work.caltech.edu/telecourse>

Second homework assignment (due date 07/10)

- Complete all coding exercises from coding session W2S1
- Term Project Team definition: work with your classmates to decide on the team members for your term project and list all team members in your team at the end of your homework's notebook

Term Project Proposal Guidelines

- Teams of 3 or 4 students will be formed by next class (07/10)
- Each team to consider the two project ideas presented in the first homework and decide on the final idea to be pursued as term project
- The project topic must be related to the topics discussed in class
- Terms projects will be graded based on:
 - Project proposal (5%): to be presented on 07/24
 - Final project presentation (5%): to be presented on 08/22
 - Final report (5%): to be submitted on 08/22

Important: if you base your term project on an existing one, you must clearly explain what new contributions/innovations are you adding in your own project

Term Project Proposal Guidelines

- Project Proposal Presentations on 07/24:
 - Introduction: What is your project about? What is the problem you are trying to solve? Any previous related works?
 - Motivation: Why you want to work on such problem? What is the impact/benefit of it? Who benefits from it?
 - Work plan: What datasets are you going to use? What types of models are you considering? What evaluation metrics are you going to use? What baseline systems are you using as reference?
 - Risk analysis: What can go wrong in your project? Any risk mitigation strategy in mind?



Remember:

Office Hours

Tuesday 07/08 6:00-7:00pm

Next Class

Thursday 07/10, 5:30-9:30pm

Second Homework Due

Thursday 07/10, 11:59pm

First Quiz

Thursday 07/10, 6:00-6:30pm